

Applied Comparative Genomics

Michael Schatz

August 31, 2026

Lecture 1: Course Overview



Welcome!

The primary goal of the course is for students to be grounded in theory and leave the course empowered to conduct independent genomic analyses.

- We will study the leading computational and quantitative approaches for comparing and analyzing genomes starting from raw sequencing data.
- The course will focus on human genomics and human medical applications, but the techniques will be broadly applicable across the tree of life.
- The topics will include (pan)-genome assembly & comparative genomics, variant identification & analysis, gene expression & regulation, personal genome analysis, and cancer genomics.

Course Webpage: <https://github.com/schatzlab/appliedgenomics2026>

Course Discussions: <https://piazza.com/jhu/fall2026/600449600649>

Class Hours: Mon + Wed @ 3:00p – 4:15p, Krieger 180

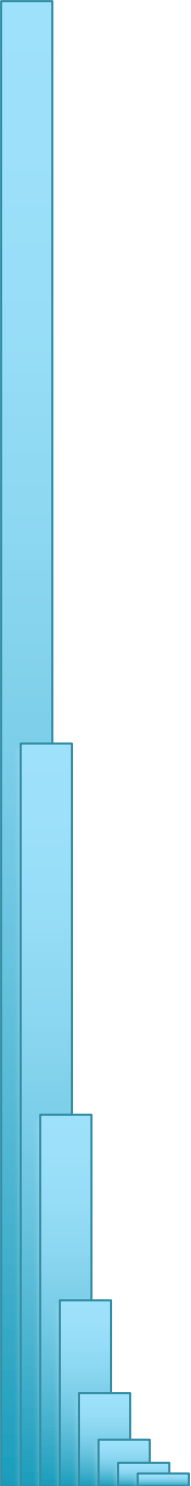
Schatz Office Hours: TBD and by appointment

TA Office Hours: TBD and by appointment

Please try Piazza first!

TA: Nicole Brown





Prerequisites and Resources

Prerequisites

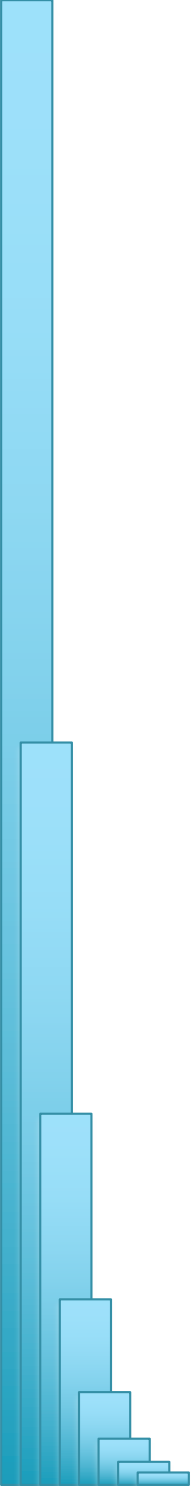
- No formal course requirements
- Access to an Apple or Linux Machine, or Install VirtualBox
- Familiarity with the Unix command line for exercises
 - bash, ls, grep, sed, + install published genomics tools
- Familiarity with a major programming language for project
 - C/C++, Java, R, Perl, Python

Primary Texts

- None! We will be studying primary research papers

Other Resources:

- Google, SEQanswers, Biostars, StackOverflow
- Applied Computational Genomics Course at UU
 - <https://github.com/quinlan-lab/applied-computational-genomics>
- Ben Langmead's teaching materials:
 - <https://www.langmead-lab.org/teaching.html>
- See course webpage



Grading Policies

Assessments:

- 5 Assignments: 25% Due at 11:59pm a week later
Practice using the tools and methods we are discussing
- 2 Exams: 40% In Class (Oct 14 & Nov 18)
Assess your performance, focusing on the concepts
- 1 Class Project: 35% Written & Oral presentations
Project Proposal 2% | Prelim report 3%
In class presentation 10 % | Written report 10 % | Oral Defense 10%
Significant project developing a novel analysis/method
- In-class Participation: Not graded, but there to help you!

Policies:

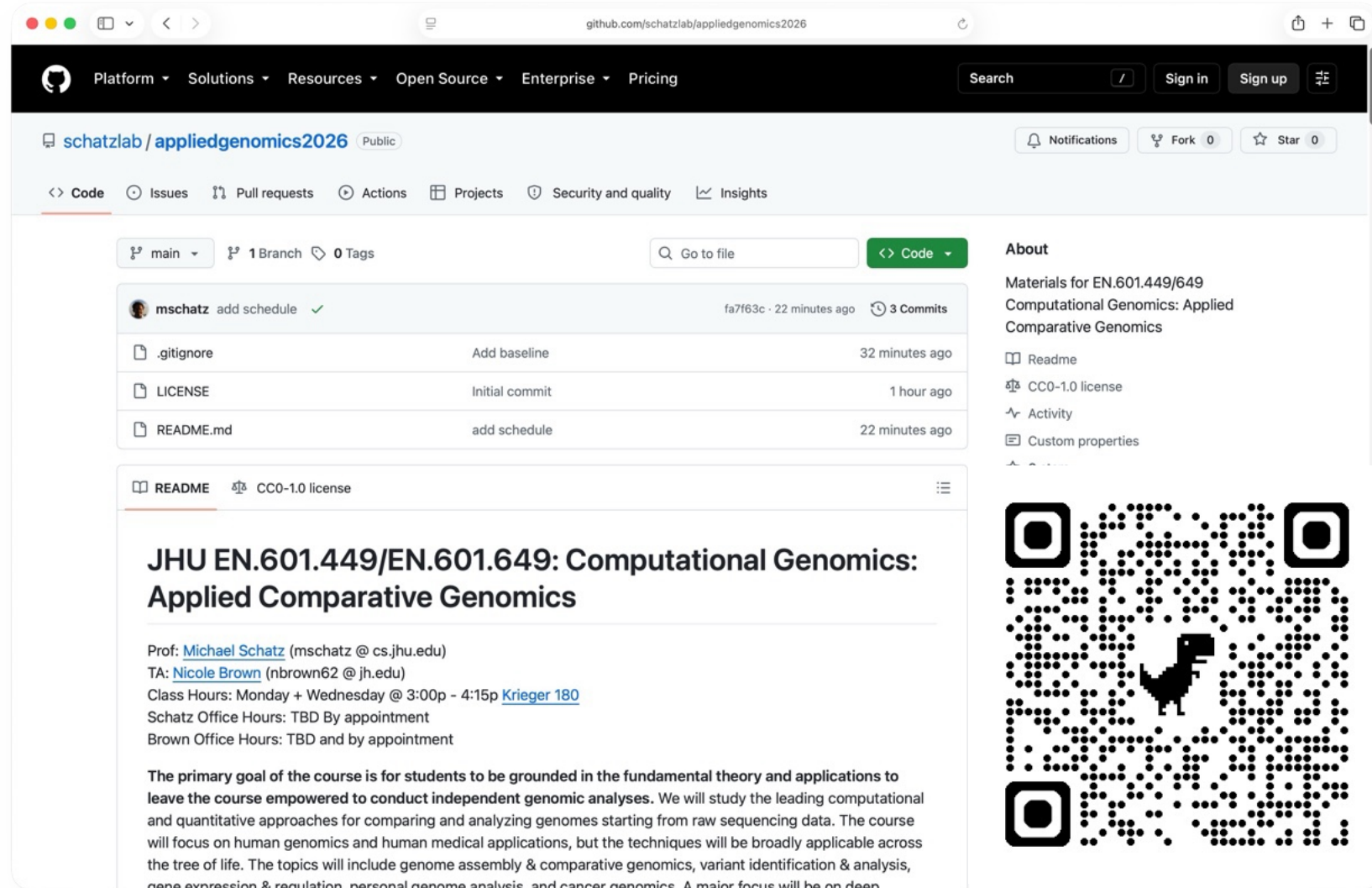
- Scores assigned relative to the highest points awarded
- Automated testing and grading of assignments
- ***Late Days:***
 - A total of 96 hours (24 x 4) can be used to extend the deadline for assignments, but not the class project, without any penalty; after that time assignments will not be accepted

On ChatGPT

whats a good policy on the use of chatgpt for a university class on genomics and data science? the instructor is generally favorable but also wants students to learn and understand the content presented in class. The instructor is also concerned that chatgpt will sometimes make up answers that are completely wrong (hallucinations)

A good policy for the use of ChatGPT in a university class on genomics and data science should emphasize that the tool is a supplement to, not a replacement for, engaging with primary course materials and learning objectives. Students should be encouraged to use ChatGPT to clarify complex concepts, find additional resources, and practice coding, but not to complete assignments or exams directly from AI-generated content without proper understanding and cross-referencing with reputable scientific sources. The policy should require students to disclose when they use ChatGPT and reflect on its contributions and limitations, fostering critical thinking and awareness of potential inaccuracies or "hallucinations." Educating students on the limitations of AI, including how to fact-check responses against trusted academic sources, will enhance their ability to use the tool responsibly. Additionally, promoting ethical use and reinforcing academic integrity are crucial, with the instructor modeling appropriate use and providing clear examples of acceptable and unacceptable practices. This approach will help students develop essential skills in genomics and data science while making the most of AI tools like ChatGPT.

Course Webpage



Platform Solutions Resources Open Source Enterprise Pricing Search Sign in Sign up

schatzlab / appliedgenomics2026 Public Notifications Fork 0 Star 0

<> Code Issues Pull requests Actions Projects Security and quality Insights

main 1 Branch 0 Tags Go to file Code

mschatz	add schedule ✓	fa7f63c · 22 minutes ago	3 Commits
.gitignore	Add baseline	32 minutes ago	
LICENSE	Initial commit	1 hour ago	
README.md	add schedule	22 minutes ago	

README CC0-1.0 license

JHU EN.601.449/EN.601.649: Computational Genomics: Applied Comparative Genomics

Prof: [Michael Schatz](#) (mschatz@cs.jhu.edu)
TA: [Nicole Brown](#) (nbrown62@jh.edu)
Class Hours: Monday + Wednesday @ 3:00p - 4:15p [Krieger 180](#)
Schatz Office Hours: TBD By appointment
Brown Office Hours: TBD and by appointment

The primary goal of the course is for students to be grounded in the fundamental theory and applications to leave the course empowered to conduct independent genomic analyses. We will study the leading computational and quantitative approaches for comparing and analyzing genomes starting from raw sequencing data. The course will focus on human genomics and human medical applications, but the techniques will be broadly applicable across the tree of life. The topics will include genome assembly & comparative genomics, variant identification & analysis, gene expression & regulation, personal genome analysis, and cancer genomics. A major focus will be on deep

About
Materials for EN.601.449/649
Computational Genomics: Applied Comparative Genomics
Readme
CC0-1.0 license
Activity
Custom properties



<https://github.com/schatzlab/appliedgenomics2026>

Piazza

The screenshot displays the Piazza web application interface. At the top, a blue header bar contains the Piazza logo, a phone number (600.449/600.649), and navigation links for Q & A, Resources, Statistics, and Manage Class. A 'Buy a License' button and a 'Switch to contribution model' link are also present. The user's name, Michael Schatz, is shown in the top right corner. Below the header, a sidebar on the left shows a list of posts under the 'Today' filter, with the top post titled 'Welcome to Piazza!' by Michael Schatz. The main content area displays the details of this post, including the title 'Welcome to Piazza!', the author 'Michael Schatz', and the text of the post. The post content reads: 'Welcome to Piazza! We'll be conducting all class-related discussion here this term. The quicker you begin asking questions on Piazza (rather than via emails), the quicker you'll benefit from the collective knowledge of your classmates and instructors. We encourage you to ask questions when you're struggling to understand a concept—you can even do so anonymously. -Michael Schatz'. Below the post content, there are buttons for 'Edit', 'Like' (0), 'Bookmark', 'Star', and 'Share'. A section for '0 Followup Discussions' is also visible, with a text input field for composing a new discussion. The bottom of the interface shows a list of tabs for different topics: hw1, hw2, hw3, hw4, hw5, project, exam1, exam2, and general.

piozza 600.449/600.649

Q & A Resources Statistics Manage Class

Buy a License Switch to contribution model Michael Schatz

New Post Search posts...

All Posts

Today

Instr Welcome to Piazza! 10:18 PM

Students,Welcome to Piazza! We'll be conducting all class-related discussion here this term. The quicker you begin askin

Note History No history yet Disable

hw1 hw2 hw3 hw4 hw5 project exam1 exam2 general

note @6

Welcome to Piazza!

Updated 1 hour ago by Michael Schatz

Students,

Welcome to Piazza! We'll be conducting all class-related discussion here this term. The quicker you begin asking questions on Piazza (rather than via emails), the quicker you'll benefit from the collective knowledge of your classmates and instructors. We encourage you to ask questions when you're struggling to understand a concept—you can even do so anonymously.

-Michael Schatz

Edit 0

0 views

0 Followup Discussions

Compose a new followup discussion

<https://piazza.com/jhu/fall2026/600449600649>

Deep Learning Reading Group

Jake Galvin will lead the discussion on the paper:
“Generative design of bacteriophages with genome language models”
<https://www.science.org/doi/10.1126/science.aec2657>

This paper went viral when it was released a few weeks ago
<https://bit.ly/4ydy6UM>

We look forward to seeing everyone there!

Meeting info:

Date: Tuesday, September 1st
Time: 12:00 p.m.–1:00 p.m.
Room: Malone Hall 338 & Zoom
Zoom link: <https://bit.ly/4ybogmf>

#deep-learning-reading-group on
JHU Genomics Collective Slack

RESEARCH ARTICLE SUMMARY

SYNTHETIC BIOLOGY

Generative design of bacteriophages with genome language models

Samuel H. King, Claudia L. Driscoll, David B. Li, Daniel Guo, Aditi T. Merchant, Garyk Brixi, Max E. Wilkinson, Brian L. Hie*

INTRODUCTION: Evolution continuously forges new biological innovations written in genomes. Navigating this vast design space could access functions that would transform biotechnology, but even the simplest genomes are highly complex and can be rendered nonviable by a single mutation. Accordingly, most progress in biological design has been made at the scale of individual genes and gene circuits, whereas designs at the scale of whole genomes has remained largely beyond reach.

RATIONALE: Genome language models are artificial intelligence (AI) algorithms that have shown promise in designing biological systems. Much like how other language models are trained on large corpora of text, genome language models are trained on large corpora of DNA comprising millions of genomes from all domains of life. This enables these models to learn the evolutionary constraints that shape DNA sequences in nature. However, the ability of genome language models to generate entire functional genomes has not been tested. Bacteriophages, viruses that infect bacteria, are specifically well suited for this task, as they are relatively small, experimentally tractable, and have broad applications in molecular biology, microbial engineering, and therapeutics.

RESULTS: In this work, we leveraged genome language models, Evo 1 and Evo 2, to generate complete phage genomes with realistic genetic architectures and specificity for a bacterial host, *Escherichia coli* C. Using the natural phage ΦX174 as a design template, we established a framework for generating and evaluating thousands of AI-generated genomes, nearly 300 of which we chemically synthesized and tested in laboratory conditions,

yielding 16 viable phages. The viable generated phages showed strong host specificity and diverse fitness profiles, including competitive infection kinetics. The generated phages were different from any known natural phages, exhibiting de novo mutations, divergent genes and regulatory elements, and variable genome lengths. One of the phages utilized a DNA packaging protein from an evolutionarily distant phage in its capsid structure. We also tested whether the generated phages could overcome bacterial resistance, a central challenge in developing phage-based antimicrobial therapies, and found that a mixture of designed phages rapidly overcame ΦX174-resistant *E. coli* strains, whereas a comparable mixture of naturally sourced ΦX174-like phages could not.

CONCLUSION: Our results demonstrate that generative models capture evolutionary constraints in DNA sequences with enough fidelity to produce complete bacteriophage genomes divergent from those observed in nature and with prespecified traits. Our approach expands what synthetic genomics can achieve alongside methods such as directed evolution and rational engineering, lays out a path for generating adaptive and resilient phage therapies against rapidly evolving pathogens, and establishes a foundation for the generative design of larger, more complex genomes. Genome design can augment the broader toolkit of genome sequencing, synthesis, and editing, enabling the composition of biological systems at the genome scale. □

*Corresponding author: Email: brianhie@stanford.edu Cite this article as S. H. King et al., Science 393, eae2657 (2026). DOI: 10.1126/science.aec2657

A framework for AI-guided bacteriophage genome design. Language models trained on millions of genomes enable the generation of complete phages. Sequences are generated and computationally evaluated with design criteria inspired from the phage ΦX174 and its host *E. coli* C. The most promising designs are chemically synthesized and tested in host cells, yielding viable generated phage genomes divergent from known natural phages, while retaining host specificity.

AI-guided bacteriophage genome design

Template: ΦX174
Host: *E. coli* C

Generate sequences

Evaluate designs

Construct genomes

Test in host cells

Modeling framework

Target host specificity

Genomic diversity

Phenotypic diversity

Evolutionary resilience

Science 6 AUGUST 2026

589

GradeScope

The screenshot shows the GradeScope interface for a course. The browser address bar displays www.gradescope.com/courses/1370921. The course title is **EN.601.449/EN.601.649** for **Fall 2026**, with an **Entry Code: 7B4VJ6**. The course ID is 1370921. The description is 'Applied Genomics'. The 'Things To Do' section lists two tasks: 'Add students or staff to your course from the [Roster](#) page.' and 'Create your first assignment from the [Assignments](#) page.' A table with columns for 'Active Assignments', 'Released', 'Due (EDT)', 'Submissions', '% Graded', 'Published', and 'Regrades' is present, but it is empty. A message states 'You currently have no assignments. Create an assignment to get started.' with a 'Create Assignment' button. The sidebar includes the GradeScope logo, a menu icon, the course title, and navigation links for 'Dashboard', 'Assignments', 'Roster', 'Extensions', and 'Course Settings'. The 'Instructor' section lists 'Michael Schatz'. The 'Course Actions' section includes 'Unenroll From Course'. The bottom of the sidebar shows an 'Account' link.

<https://www.gradescope.com/courses/1370921>

Entry Code: **7B4VJ6**

Part 1: Genome Assembly & Analysis (DNA-based)

Schedule

Class	Date	Day	Topic	Assignments	Readings
1	8/31/26	M	Introduction	Sign up for Piazza	Molecular Structure of Nucleic Acid (Watson and Crick, 1953, Nature) Biological data sciences in genome research (Schatz, 2015, Genome Research) Big Data: Astronomical or Genomical? (Stephens et al, 2015, PLOS Biology)
2	9/2/26	W	Genomics Technologies		Coming of age: ten years of next-generation sequencing technologies (Goodwin et al, 2016, Nature Reviews Genetics) Guide to k-mer approaches for genomics across the tree of life (Jenike et al., 2024, arXiv)
*	9/7/26	M	Labor Day		
3	9/9/26	W	Assembly and WGA	HW1	Toward simplifying and accurately formulating fragment assembly. (Myers, 1995, J. Comp. Bio.) Velvet: Algorithms for de novo short read assembly using de Bruijn graphs (Zerbino and Birney, 2008, Genome Research) SPAdes: A New Genome Assembly Algorithm and Its Applications to Single-Cell Sequencing (Bankevich, et al. 2012, J Comput Biol) MUMmer: Alignment of Whole Genomes (Delcher et al, 1999, NAR)
4	9/14/26	M	The Human Genome		The complete sequence of a human genome (Nurk et al, Science 2012) Approaching complete genomes, transcriptomes and epi-omes with accurate long-read sequencing (Kovaka et al, 2023, Nature Methods) A draft human pangenome reference (Liao et al, 2023, Nature) Beyond the Human Genome Project: The Age of Complete Human Genome Sequences and Pangenome References (Taylor et al., 2024, Annual Review of Genomics and Human Genetics) A complete diploid human genome benchmark for personalized genomics (Hansen et al, 2026, Cell)
5	9/16/26	W	Read Mapping	HW1 due	How to map billions of short reads onto genomes (Trapnell and Salzberg, 2009, Nature Biotech) Sapling: Accelerating Suffix Array Queries with Learned Data Models (Kirsche et al, 2020, Bioinformatics)
6	9/21/26	M	BWT	HW2	Bowtie: Ultrafast and memory-efficient alignment of short DNA sequences to the human genome (Langmead et al, 2009, Genome Biology) BWA-MEM: Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM (Li, 2013, arXiv) Minimap2: pairwise alignment for nucleotide sequences (Li, Bioinformatics, 2018)
7	9/23/26	W	Variant Analysis		Haplotype-based variant detection from short-read sequencing (Garrison and Marth, arXiv, 2012) The Genome Analysis Toolkit: A MapReduce framework for analyzing next-generation DNA sequencing data (McKenna et al, 2010, Genome Research) A universal SNP and small-indel variant caller using deep neural networks (Poplin et al, 2018, Nature Biotechnology) SAM/BAM/Samtools: The Sequence Alignment/Map format and SAMtools (Li et al, 2009, Bioinformatics) IGV: Integrative genomics viewer (Robinson et al, 2011, Nature Biotech)
8	9/28/26	M	Intro to ML	HW2 due	What are decision trees? (Kingsford and Salzberg, 2008, Nature Biotechnology) What is a hidden Markov model? (Eddy, 2004, Nature Biotechnology) Deep learning in biomedicine (Wainberg et al, 2018, Nature Biotechnology) Visualizing Data Using t-SNE
9	9/30/26	W	CNN + DeepVariant	HW3	ImageNet Classification with Deep Convolutional Neural Networks (Krizhevsky et al., 2012, NIPS) A universal SNP and small-indel variant caller using deep neural networks (Poplin et al. 2018, Nature Biotech)
10	10/5/26	M	Population Genetics		An integrated map of genetic variation from 1,092 human genomes (1000 Genomes Consortium, 2012, Nature) Analysis of protein-coding genetic variation in 60,706 humans (Let et al, 2016, Nature) A Draft Sequence of the Neandertal Genome (Green et al. 2010, Science) Excavating Neandertal and Denisovan DNA from the genomes of Melanesian individuals (Vernot et al. 2016, Science) Inverting the model of genomics data sharing with the NHGRI Genomic Data Science Analysis, Visualization, and Informatics Lab-space (AnVIL) (Schatz et al, 2022, Cell Genomics)
11	10/7/26	W	Clinical Genomics	HW3 due	Genome-Wide Association Studies (Bush & Moore, 2012, PLOS Comp Bio) The contribution of de novo coding mutations to autism spectrum disorder (Iossifov et al, 2014, Nature)
12	10/12/26	M	Review 1	Project Proposal	
13	10/14/26	W	Exam 1 (In class)		

<https://github.com/schatzlab/appliedgenomics2026>

Part 2: Functional Genomics & Deep Learning

14	10/19/26	M	Functional Analysis 1: RNA-seq	Project Proposal due	• RNA-Seq: a revolutionary tool for transcriptomics (Wang et al, 2009, Nature Reviews Genetics) • Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks (Trapnell et al, 2012, Nature Protocols) • Salmon provides fast and bias-aware quantification of transcript expression (Patro et al, 2017, Nature Methods) • Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications (Krueger and Andrews, 2011, Bioinformatics)
15	10/21/26	W	Functional Analysis 2: Single Cell Genomics		• Ginkgo: Interactive analysis and assessment of single-cell copy-number variations (Garvin et al, 2015, Nature Methods) • The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells (Trapnell et al, Nature Biotech, 2014) • Eleven grand challenges in single-cell data science (Lahnemann et al, Genome Biology, 2020)
16	10/26/26	M	Functional Analysis 3: Ab initio gene finding	HW4	• BLAST: Basic Local Alignment Search Tool • Glimmer: Microbial gene identification using interpolated Markov models • MAKER2: an annotation pipeline and genome-database management tool for second-generation genome projects • BEDTools: a flexible suite of utilities for comparing genomic features (Quinlan & Hall, 2010, Bioinformatics)
17	10/28/26	W	Functional Analysis 4: Methyl-seq, Chip-seq, and Hi-C		• ChIP-seq and beyond: new and improved methodologies to detect and characterize protein-DNA interactions (Furey, 2012, Nature Reviews Genetics) • PeakSeq enables systematic scoring of ChIP-seq experiments relative to controls (Rozowsky et al. 2009, Nature Biotech) • Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome (Lieberman-Aiden et al, 2009, Science)
18	11/2/26	M	Functional Analysis 5: Regulatory States, ENCODE, GTEx, RoadMap	HW4 due; Prelim report assigned	• An integrated encyclopedia of DNA elements in the human genome (The ENCODE Project Consortium, Nature, 2012) • Genetic effects on gene expression across human tissues (GTEx Consortium, Nature, 2017) • Integrative analysis of 111 reference human epigenomes (Roadmap Epigenome Consortium, Nature, 2015) • ChromHMM: automating chromatin-state discovery and characterization (Ernst & Kellis, 2012, Nature Methods) • Segway: Unsupervised pattern discovery in human chromatin structure through genomic segmentation (Hoffman et al, 2012, Nature Methods)
19	11/4/26	W	No class - Biological Data Sciece		
20	11/9/26	M	Transformers	HW5	• Attention is all you need (Vaswani et al. 2017, arXiv)
21	11/11/26	W	Enformer + Other DL applications		• Effective gene expression prediction from sequence by integrating long-range interactions (Avsec et al., 2021, Nature Methods) • Personal transcriptome variation is poorly explained by current genomic deep learning models (Huang et al., 2023, Nature Genetics) • Benchmarking of deep neural networks for predicting personal gene expression from DNA sequence highlights shortcomings (Sasse et al., 2023, Nature Genetics)
22	11/16/26	M	Review 2	HW5 due; Final Report Assigned	
23	11/18/26	W	Exam 2 (In class)		
*	11/23/26	M	Thanksgiving Break		
*	11/25/26	W	Thanksgiving Break		
24	11/30/26	M	Wrap up		• Deep Learning Sequence Models for Transcriptional Regulation (Sokolova et al., 2024, Annual Reviews of Genomics and Human Genetics) • AlphaFold (Jumper et al, 2021, Nature)

<https://github.com/schatzlab/appliedgenomics2026>

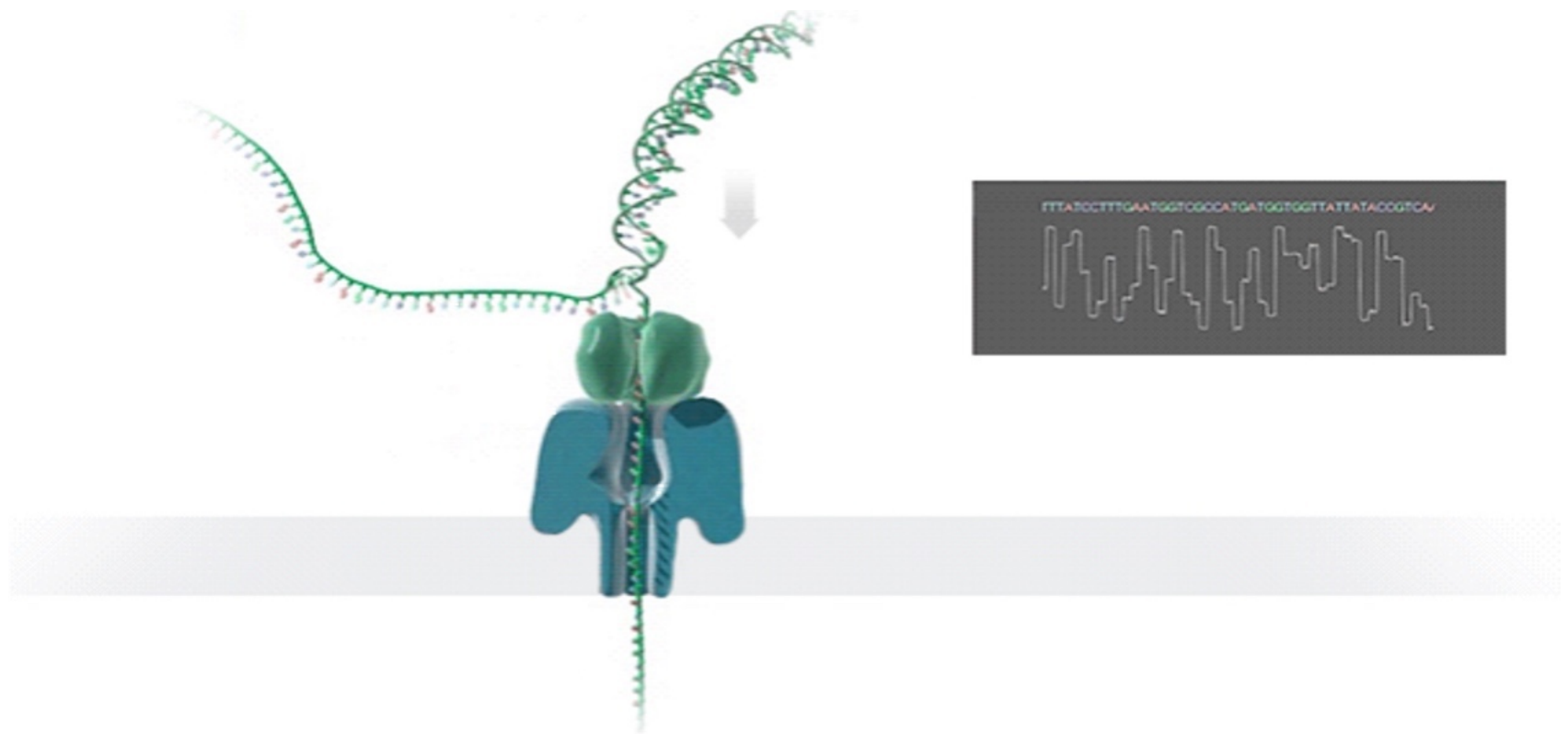
Part 3: Class Projects!

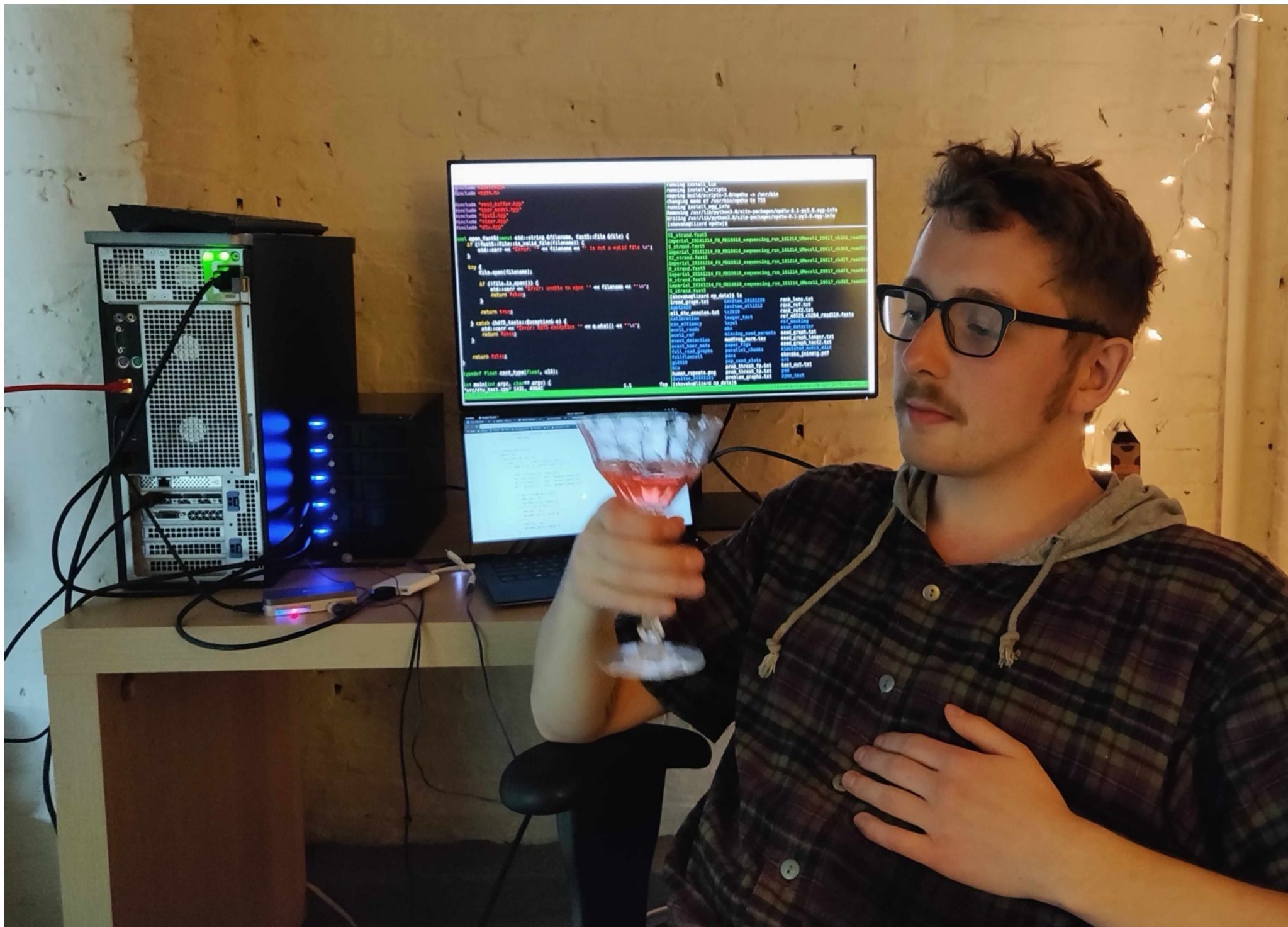
24	11/30/26	M	Wrap up		* Deep Learning Sequence Models for Transcriptional Regulation (Sokolova et al., 2024, Annual Reviews of Genomics and Human Genetics) * AlphaFold (Jumper et al, 2021, Nature)
25	12/2/26	W	In-class presentation	In-class presentation	
26	12/7/26	M	In-class presentation	In-class presentation	
27	12/9/26	W	In-class presentation	In-class presentation	
*	12/13/26	S	Draft Report Due	Draft Report Due	
*	12/14/26	M	Final project presentation	Final project presentation	
*	12/15/26	Tu	Final project presentation	Final project presentation	
*	12/16/26	W	Final project presentation	Final project presentation	
*	12/17/26	Th	Final Report Due	Final Report Due	

<https://github.com/schatzlab/appliedgenomics2026>

Nanopore Sequencing

Sequences DNA/RNA by measuring changes in ionic current as nucleotide strand passes through a pore





Subject Section

Basecall-free alignment of Oxford Nanopore reads using the Burrow-Wheelers Transform

Yunfan Fan^{1,*}, Sam Kovaka² and Taher Mun^{2,*}

¹Department of Biomedical Engineering, Johns Hopkins University, Baltimore, 21218, USA and

²Department of Computer Science, Johns Hopkins University, Baltimore, USA.

*To whom correspondence should be addressed.

pre-print

Abstract

Motivation: Sequencers manufactured by Oxford Nanopore Technologies (ONT) can generate long reads while simultaneously providing the ability to sequence in the field due to the compact cell-phone-sized design of their MinION sequencer. However, the technology hinges on measuring minute changes in ionic current through a pore as DNA is simultaneously passed through. The process of basecalling, which converts these electrical signals into bases, is a time-consuming and resource intensive process, but necessary for most downstream applications such as read alignment and genome assembly. This project aims to bypass the basecalling step for alignment applications by directly aligning the signals to a reference genome that has been converted from bases into signals.

Results: results here

Availability: availability here

Contact: yfan7@jhu.edu, skovaka1@jhu.edu, tmun1@jhu.edu

Supplementary information: Available upon request.

1 Introduction

Oxford Nanopore reads provide the convenience of portability, the immediacy of real-time sequencing, as well as the novelty and usefulness of long reads. Since the ONT MinION technology is as compact as a cell-phone and connects directly to a conventional laptop via USB to operate, it has been leveraged in the field for applications including outbreak monitoring (Quick et al. 2016) and extraterrestrial sequencing (Rainey 2016). Its capability to gather and report useful data in real time has also inspired efforts to apply DNA sequencing in urgent medical diagnosis settings (Votintseva et al. 2017).

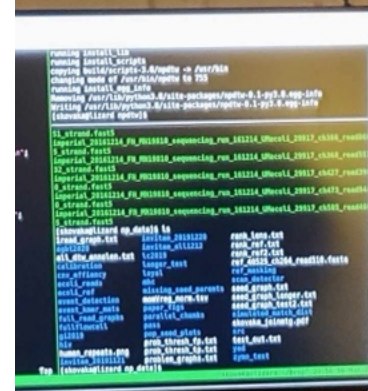
The MinION sequencer measures and records a time series of electrical current amplitudes as ions and a DNA strand pass through a nanometer sized hole. As segments of six base pairs occupy the pore sequentially, each recorded current corresponds to unique 6-mers of DNA. Due to this indirect nature of sequencing, the actual base sequence of the reads need to be eluted through a process called basecalling, where signals are translated into nucleotides. Base calling software and services are available for hire through ONT subsidiaries, but open source implementations are also available. However, the base calling process is invariably expensive computationally, and involves complicated modeling by way

of Hidden Markov Models in the case of the open source software or Recurrent Neural Networks in the case of the proprietary. Previously, the proprietary basecalling software also required an internet connection, which can be prohibitive of field work applications in remote locations. As most downstream software for lynchpin genomic analysis such as read alignment and genome assembly deal only with nucleotide sequences, the basecalling step is emerging as a critical bottleneck for the efficacy of the technology.

In order to address this problem in the context of read alignment, we propose to skip the base calling step altogether and align the raw current signals directly to a reference (Laszlo et al. 2014). This can be done by converting a reference genome from base-space to signal-space using a model of expected current levels and then indexing it using the Burrows-Wheeler Transform. Then, the signals generated from the ONT reads can be directly aligned to the reference using this BWT in order to obtain an accurate mapping.

2 Methods

Resources: Development and testing was done on the MARCC langmead-bismem server, which has 1TB RAM and 112 cores. Of this, we used up to 30GB RAM and a single core.





Targeted nanopore sequencing x +

← → ↺ ⌂

nature.com/articles/s41587-020-0731-9

☆

JHUMail Daily s j P I GRANTS jhu Media Rm Cookies james shop edit

» Other Bookmarks

nature biotechnology

View all Nature Research journals

Search My Account

Explore content Journal information Publish with us Subscribe

Sign up for alerts RSS feed

nature > nature biotechnology > articles > article

Article | Published: 30 November 2020

Targeted nanopore sequencing by real-time mapping of raw electrical signal with UNCALLED

Sam Kovaka , Yunfan Fan, Bohan Ni, Winston Timp & Michael C. Schatz

Nature Biotechnology (2020) | Cite this article

5715 Accesses | 2 Citations | 261 Altmetric | Metrics

Abstract

Conventional targeted sequencing methods eliminate many of the benefits of nanopore sequencing, such as the ability to accurately detect structural variants or epigenetic modifications. The ReadUntil method allows nanopore devices to selectively eject reads from pores in real time, which could enable purely computational targeted sequencing. However, this requires rapid identification of on-target reads while most mapping methods require computationally intensive basecalling. We present UNCALLED (<https://github.com/skovaka/UNCALLED>), an open source mapper that rapidly matches streaming of nanopore current signals to a reference sequence. UNCALLED probabilistically

You have full access to this article via **Johns Hopkins Libraries**

Download PDF

Sections

Figures

References

Abstract

Main

Results

Discussion

Methods

Data availability

Code availability

References

Acknowledgements

Author information

Ethics declarations

Targeted nanopore sequencing

nature.com/articles/s41587-020-0731-9

☆

Other Bookmarks

nature biotechnology

View all Nature Research journals

Search

My Account

nature methods

View all journals

Search

Log in

Sign up for alerts

RSS feed

nature > nature methods > articles > article

Article | [Open access](#) | Published: 28 March 2025

Uncalled4 improves nanopore DNA and RNA modification detection via fast and accurate signal alignment

[Sam Kovaka](#) [Paul W. Hook](#), [Katharine M. Jenike](#), [Vikram Shivakumar](#), [Luke B. Morina](#), [Roham Razaghi](#), [Winston Timp](#) & [Michael C. Schatz](#)

[Nature Methods](#) **22**, 681–691 (2025) | [Cite this article](#)

17k Accesses | 12 Citations | 55 Altmetric | [Metrics](#)

Abstract

Nanopore signal analysis enables detection of nucleotide modifications from native DNA and RNA sequencing, providing both accurate genetic or transcriptomic and epigenetic information without additional library preparation. At present, only a limited set of modifications can be directly basecalled (for example, 5-methylcytosine), while most others require exploratory methods that often begin with alignment of nanopore signal to a nucleotide reference. We present Uncalled4, a toolkit for nanopore signal alignment, analysis and visualization. Uncalled4 features an efficient banded signal alignment algorithm, BAM signal alignment file format, statistics for comparing signal alignment methods and a reproducible de novo training method for *k*-mer-based pore models, revealing potential errors in Oxford Nanopore Technologies' state-of-the-art DNA model. We apply Uncalled4 to RNA 6-methyladenine (m6A) detection in seven human cell lines, identifying 26% more modifications than Nanopolish using m6Anet, including in several genes where m6A has known implications in cancer. Uncalled4 is available open source at github.com/skovaka/uncalled4.

Download PDF

Sections

Figures

References

[Abstract](#)

[Main](#)

[Results](#)

[Discussion](#)

[Methods](#)

[Data availability](#)

[Code availability](#)

[References](#)

[Acknowledgements](#)

[Author information](#)

[Ethics declarations](#)

[Peer review](#)

[Additional information](#)

[Extended data](#)

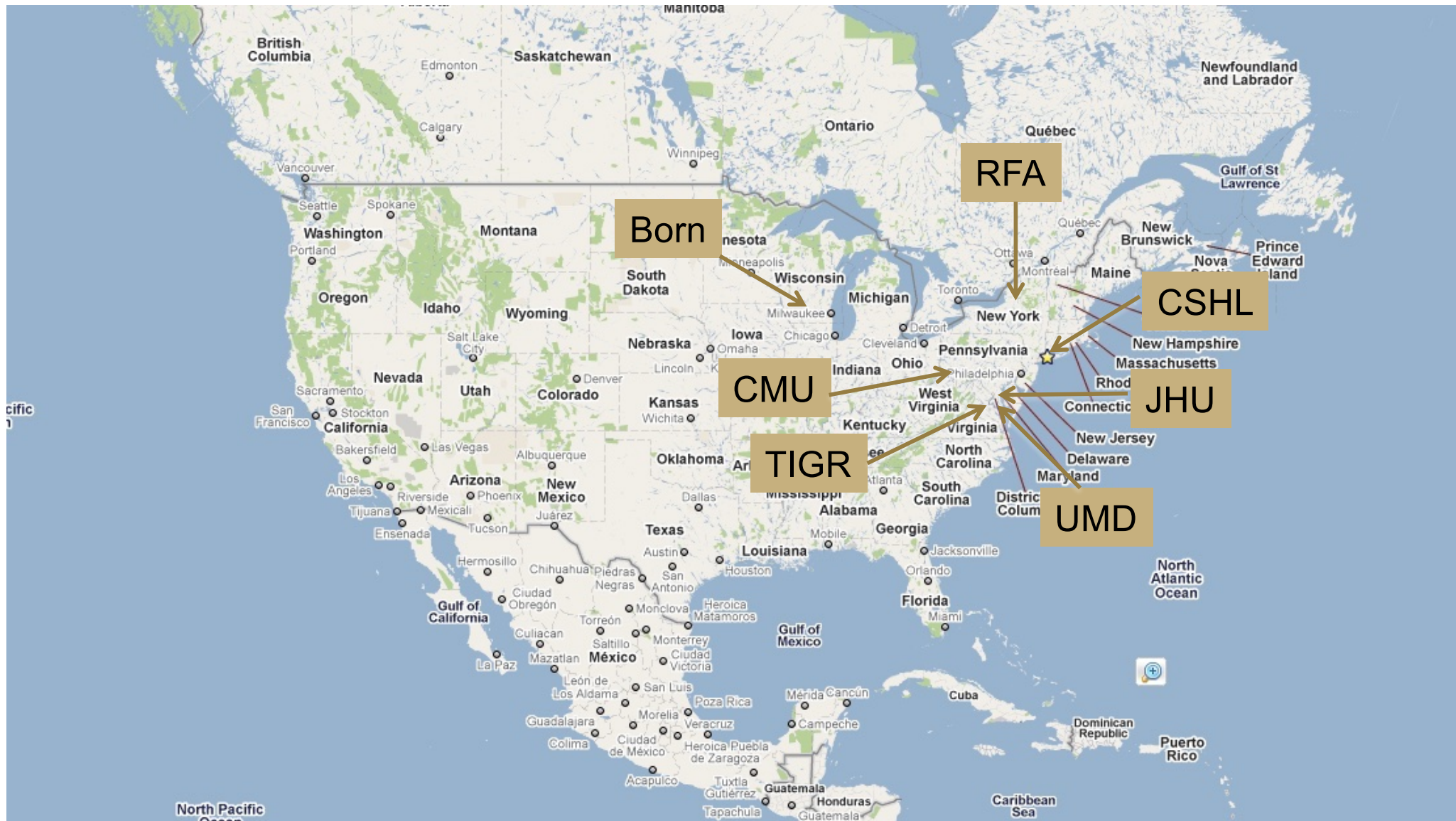
[Supplementary information](#)

[Rights and permissions](#)

[About this article](#)

[This article is cited by](#)

A Little About Me



Schatzlab Overview



Human Genetics

Role of mutations
in disease

Nurk *et al.* (2022)
Aganezov *et al.* (2020)



Agricultural Genomics

Genomes &
Transcriptomes

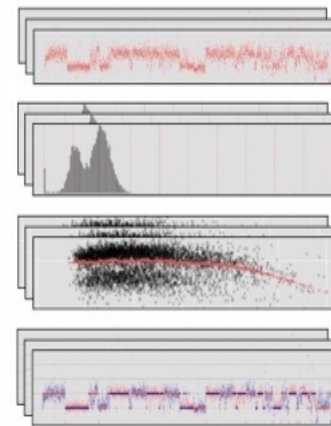
Benoit *et al.* (2025)
Satterlee *et al.* (2024)



Algorithmics & Systems Research

Ultra-large scale
biocomputing

Kirsche *et al.* (2023)
Schatz *et al.* (2022)



Biotechnology Development

Single Cell + Single
Molecule Sequencing

Kovaka *et al.* (2024)
Rozowsky *et al.* (2023)

Why Genomics?

Discovery of the Double Helix

No. 4356 April 25, 1953

NATURE

737

equipment, and to Dr. G. E. R. Deacon and the captain and officers of R.R.S. *Discovery II* for their part in making the observations.

¹Young, P. B., Gerrard, H., and Jevons, W., *Phil. Mag.*, **40**, 149 (1925).

²Longuet-Higgins, M. S., *Mon. Not. Roy. Astro. Soc., Geophys. Supp.*, **5**, 285 (1949).

³Von Arx, W. S., *Woods Hole Papers in Phys. Oceanogr. Meteor.*, **11** (3) (1950).

⁴Ekman, V. W., *Arkiv. Mat. Astron. Fysik. (Stockholm)*, **2** (11) (1905).

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey¹. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons: (1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions, namely, that each chain consists of phosphate diester groups joining β -D-deoxy-ribofuranose residues with 3',5' linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow right-handed helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions. Each chain loosely resembles Furbert's² model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furbert's 'standard configuration', the sugar being roughly perpendicular to the attached base. There



This figure is purely diagrammatic. The two ribbons symbolize the two phosphate-sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis.

is a residue on each chain every 3.4 Å. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 Å. The distance of a phosphorus atom from the fibre axis is 10 Å. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-coordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain is given, then the sequence on the other chain is automatically determined.

It has been found experimentally^{3,4} that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid.

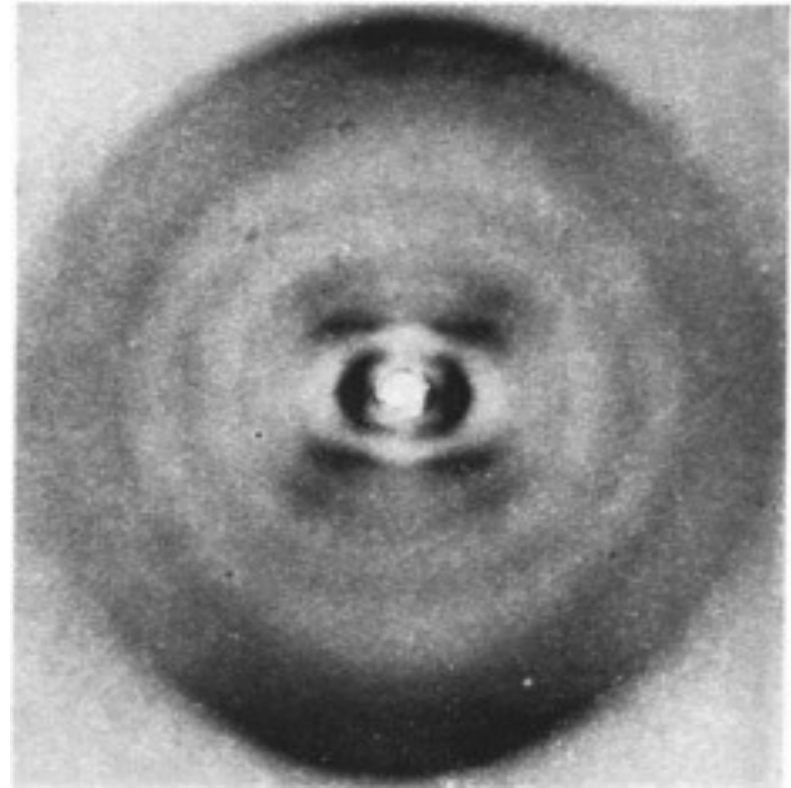
It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data^{5,6} on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in the following communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereochemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their co-workers at



CRICK AND WATSON

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material. Full details of the structure, including the con-

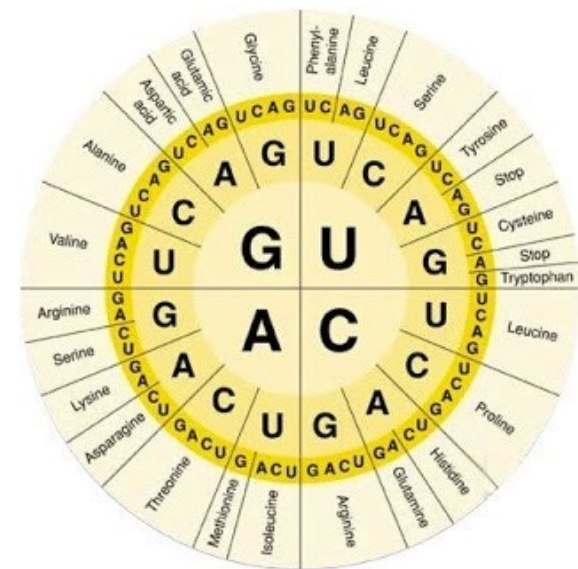
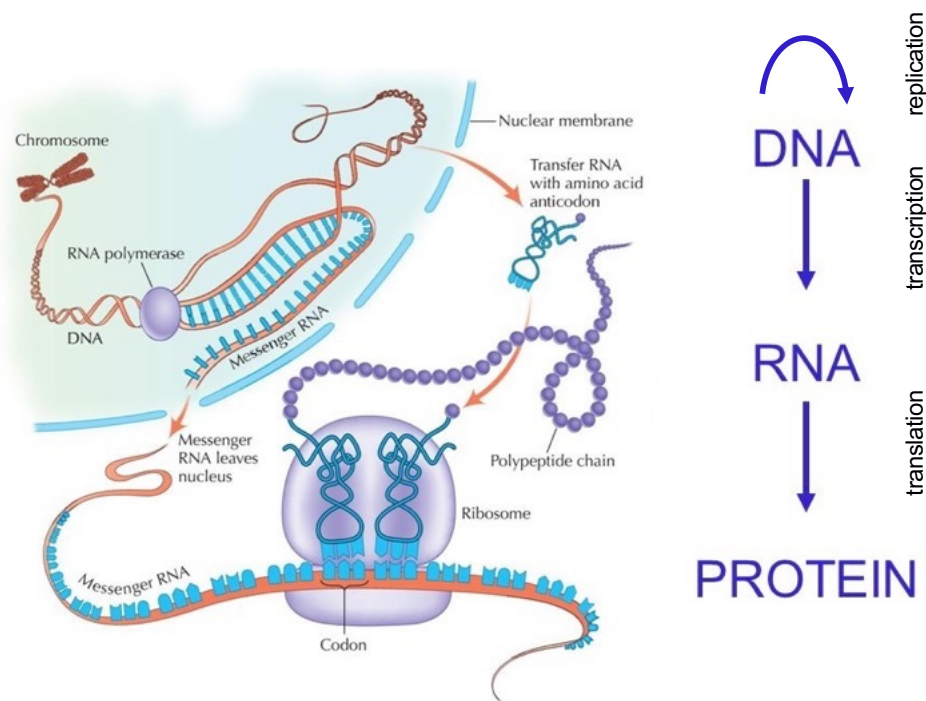
Molecular structure of nucleic acids; a structure for deoxyribose nucleic acid

Watson JD, Crick FH (1953). *Nature* 171: 737–738.

Nobel Prize in Physiology or Medicine in 1962

Central Dogma of Molecular Biology

“Once 'information' has passed into protein it cannot get out again. In more detail, the transfer of information **from nucleic acid to nucleic acid, or from nucleic acid to protein may be possible**, but transfer from protein to protein, or from protein to nucleic acid is impossible. Information means here the precise determination of sequence, either of bases in the nucleic acid or of amino acid residues in the protein”

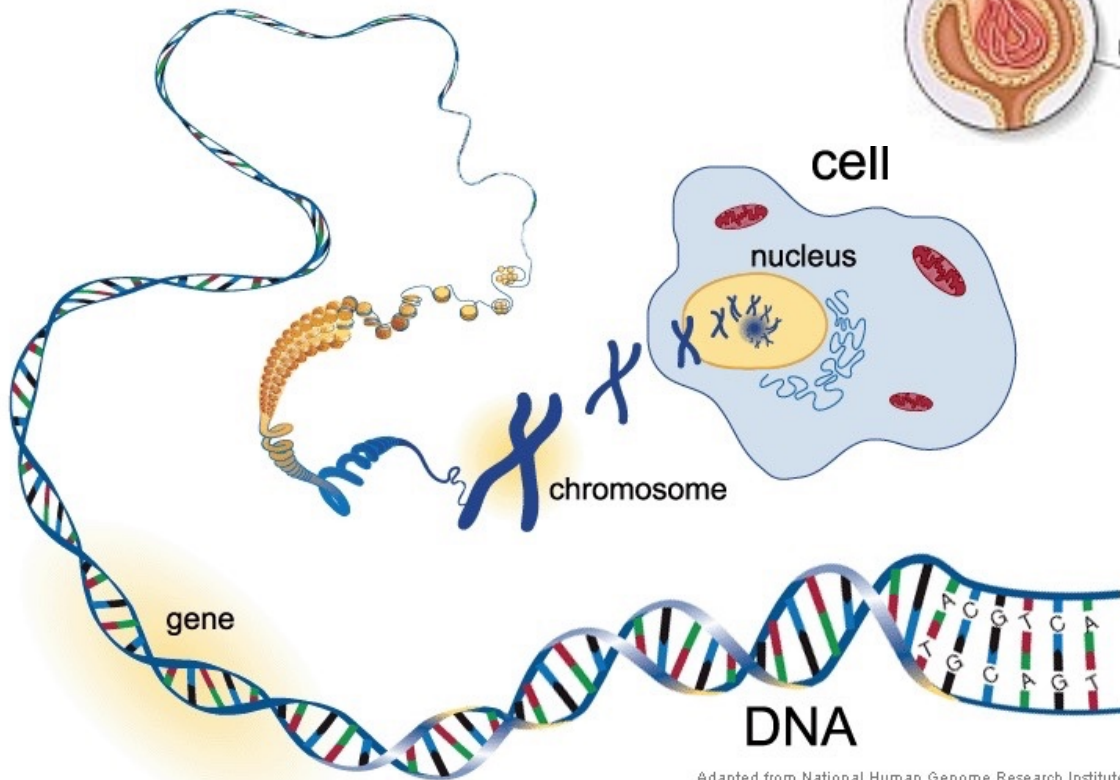
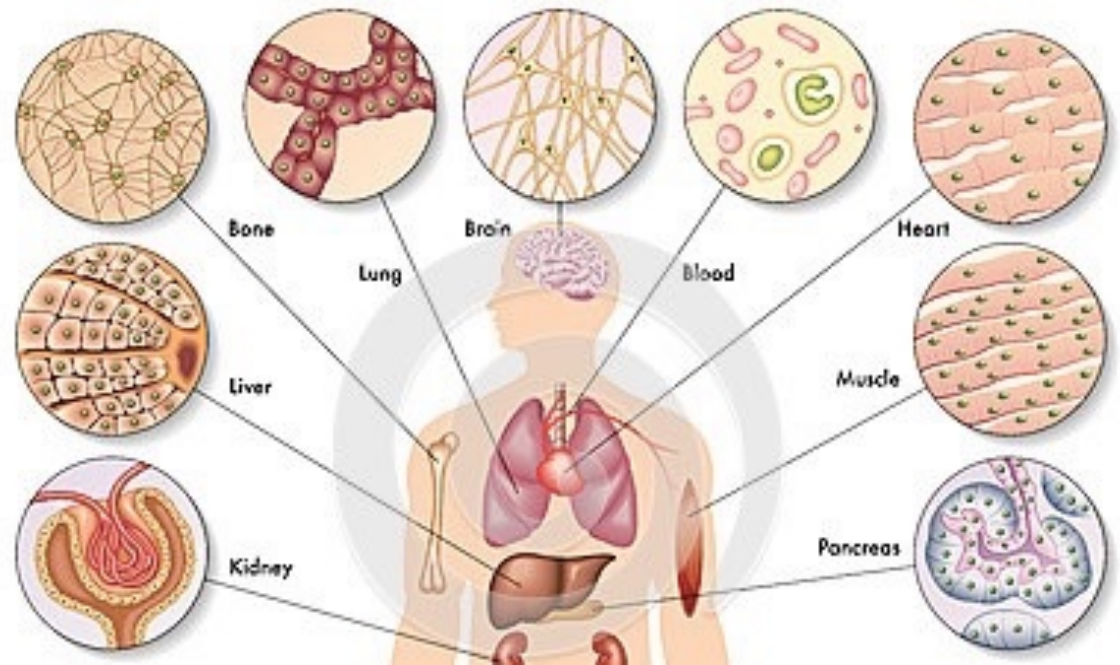


On Protein Synthesis

Crick, F.H.C. (1958). Symposia of the Society for Experimental Biology pp. 138–163.

One Genome, Many Cell Types

Each cell of your body contains an exact copy of your 3 billion base pair genome.



Your body has a few hundred (thousands?) major cell types, largely defined by the gene expression patterns

Unsolved Questions in Biology

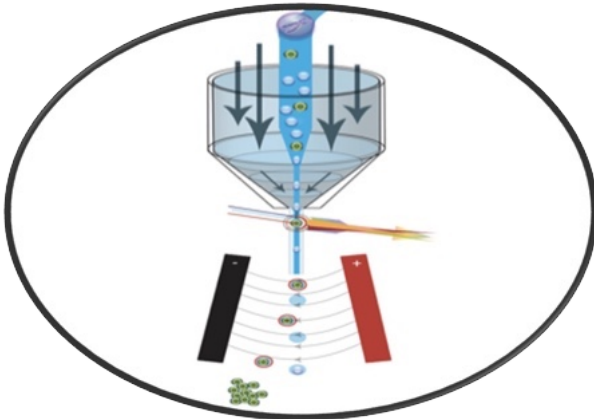
- What is your genome sequence?
- How does your genome compare to my genome?
- Where are the genes and how active are they?
- How does gene activity change during development?
- How does splicing change during development?
- How does methylation change during development?
- How does chromatin change during development?
- How does is your genome folded in the cell?
- Where do proteins bind and regulate genes?
- What virus and microbes are living inside you?
- How do your mutations relate to disease?
- What drugs and treatments should we give you?
- ***Plus thousands and thousands more***



How to do Genomics?

Genomics Arsenal in the year 2026

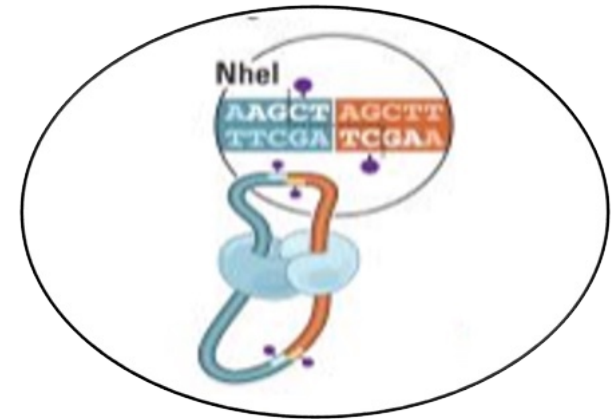
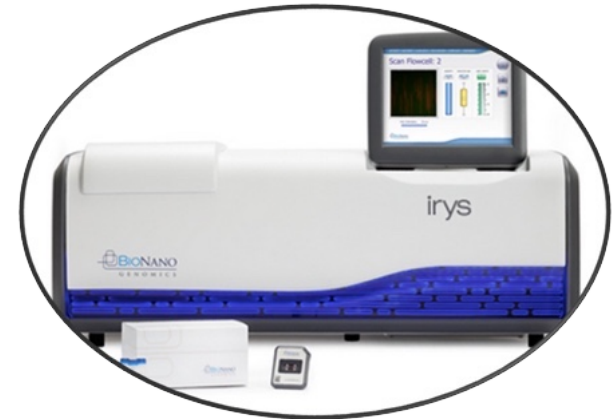
Sample Preparation



Sequencing



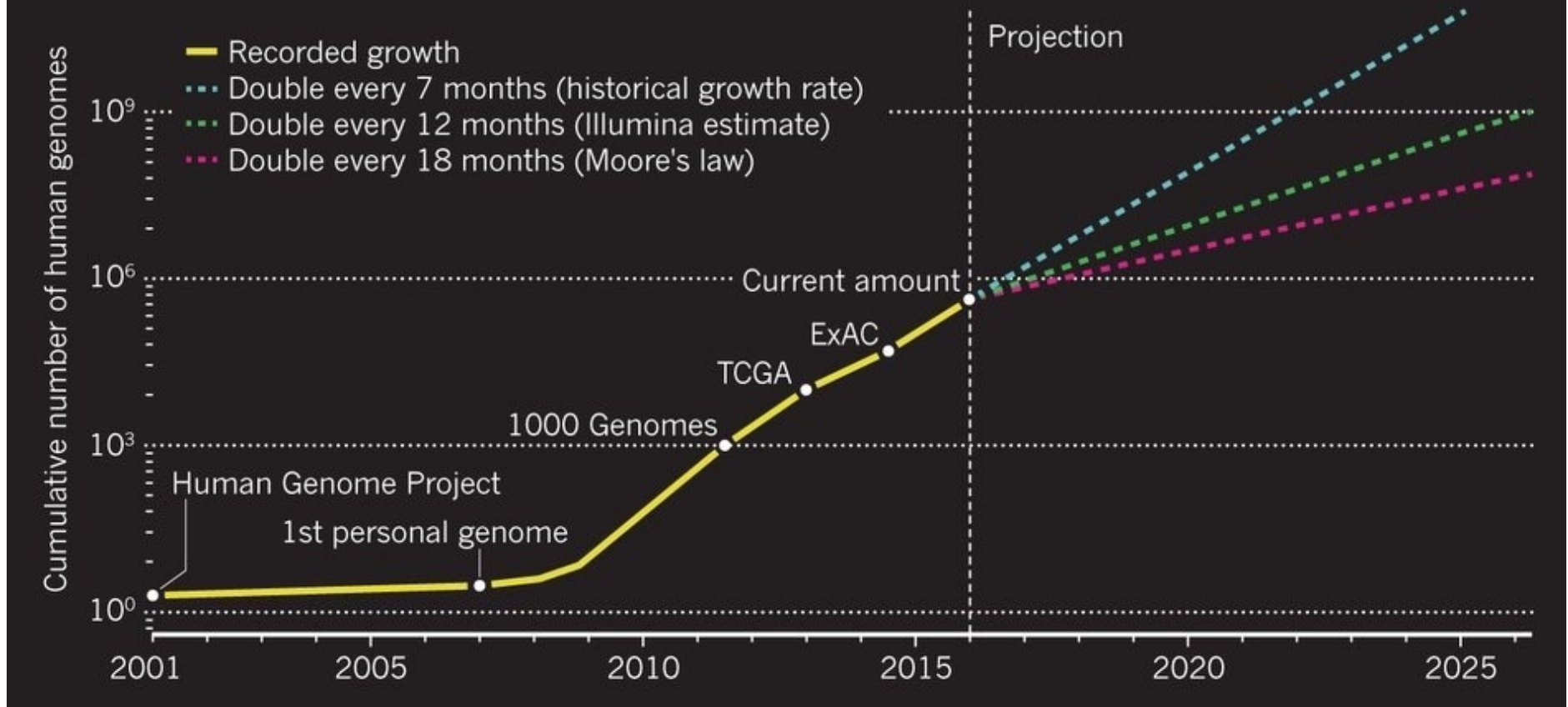
Chromosome Mapping



Sequencing Capacity

DNA SEQUENCING SOARS

Human genomes are being sequenced at an ever-increasing rate. The 1000 Genomes Project has aggregated hundreds of genomes; The Cancer Genome Atlas (TCGA) has gathered several thousand; and the Exome Aggregation Consortium (ExAC) has sequenced more than 60,000 exomes. Dotted lines show three possible future growth curves.



Big Data: Astronomical or Genomical?

Stephens, Z, et al. (2015) PLOS Biology DOI: [10.1371/journal.pbio.1002195](https://doi.org/10.1371/journal.pbio.1002195)

Sequencing Capacity

DNA SEQUENCING SOARS

Human
aggreg
the Ex
three p

Cumulative number of human genomes

10^9

10^6

10^3

10^0

2001

2005

2010

2015

2020

2025

The instruments provide the data, but none of the answers to any of these questions.

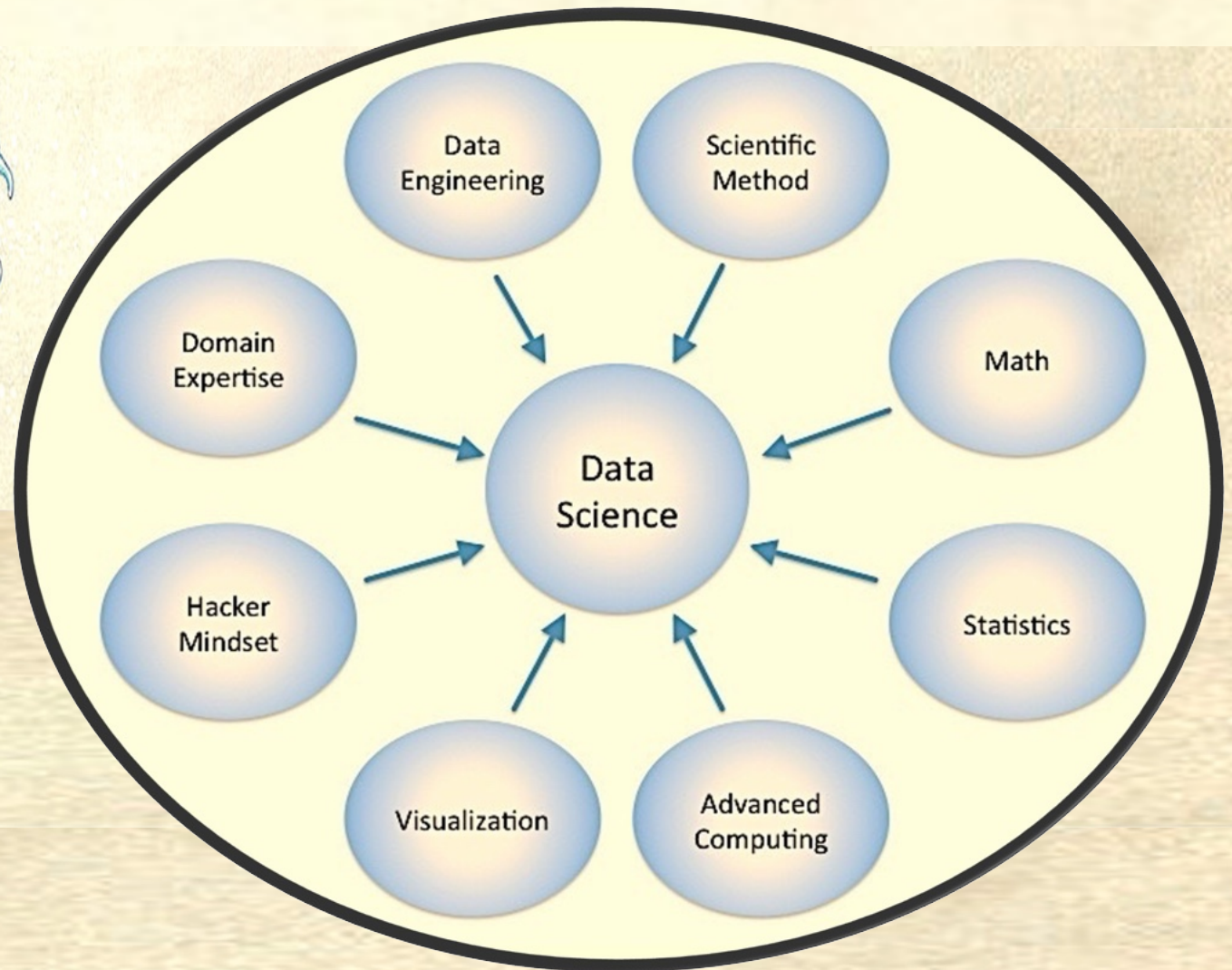
What software and systems will?

And who will create them?

Big Data: Astronomical or Genomical?

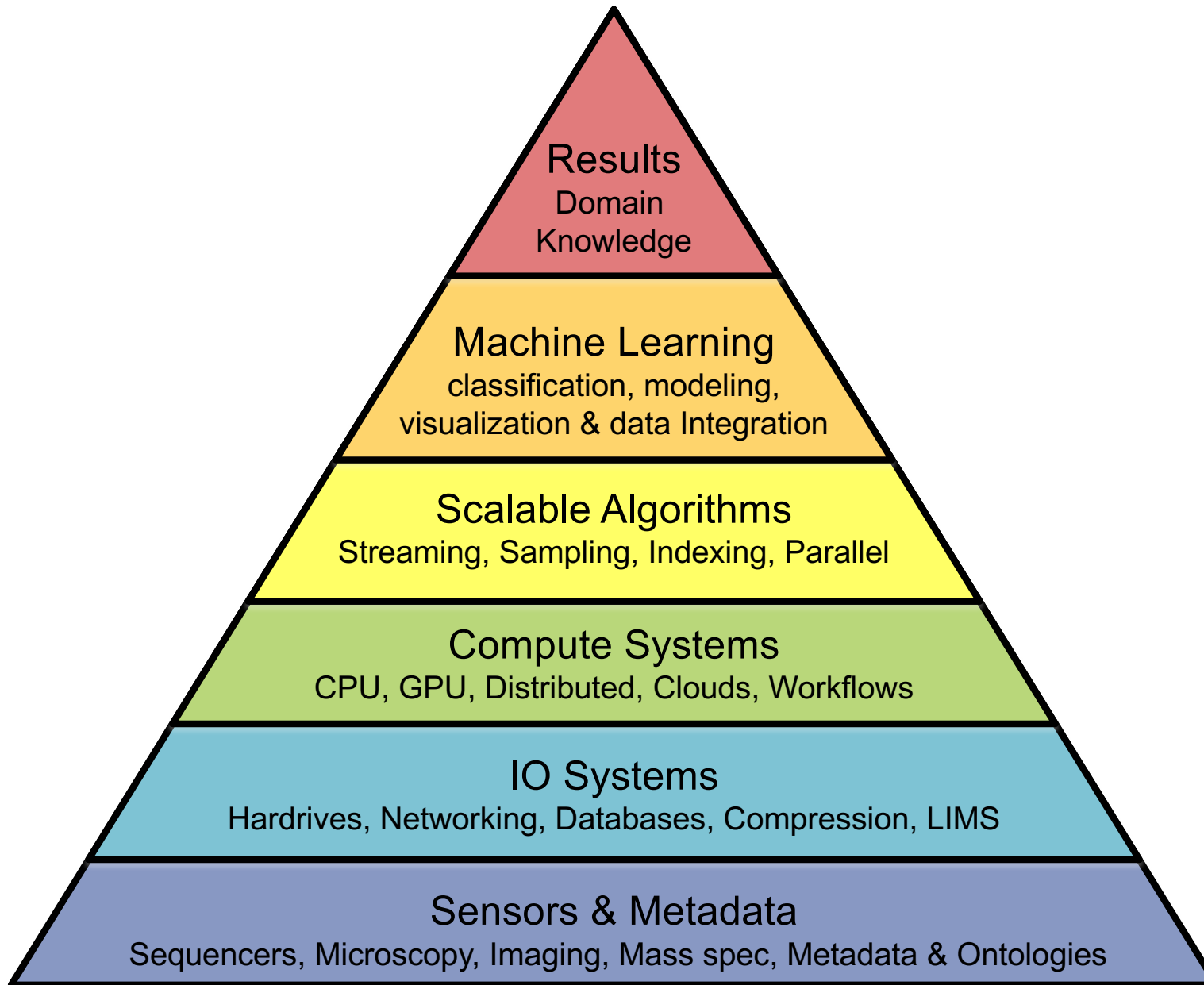
Stephens, Z, et al. (2015) PLOS Biology DOI: 10.1371/journal.pbio.1002195

Who is a Data Scientist?

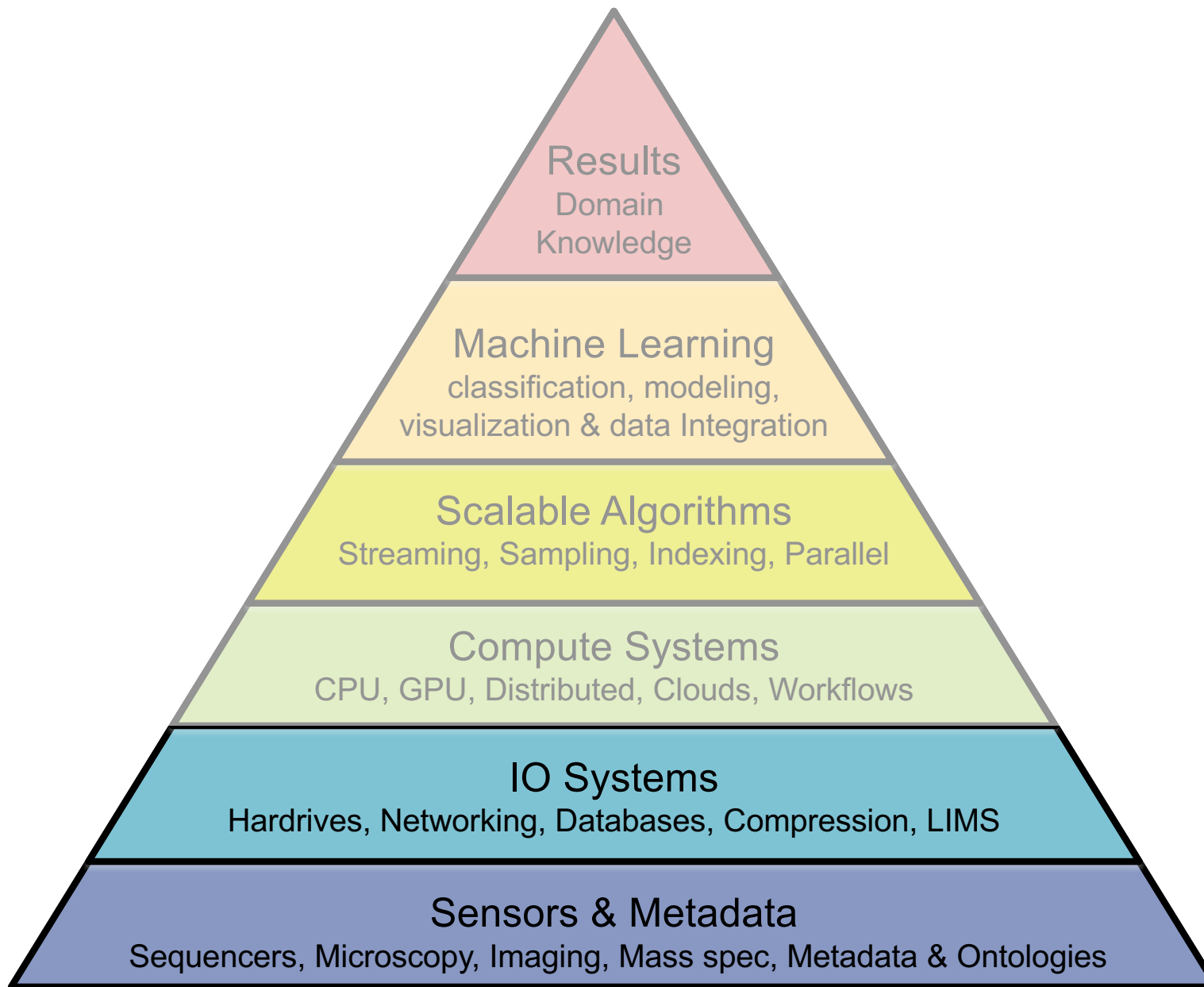


http://en.wikipedia.org/wiki/Data_science

Applied Genomics

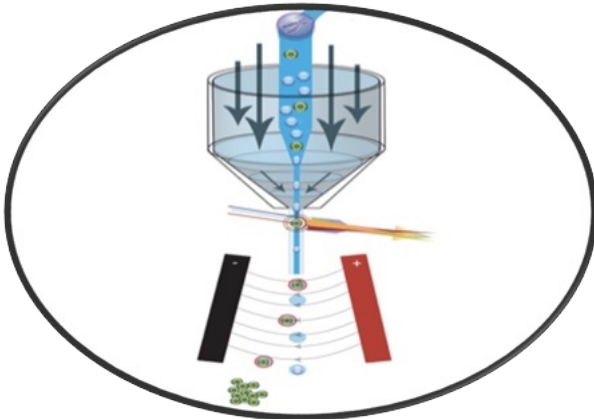


Applied Genomics



Genomics Arsenal in the year 2026

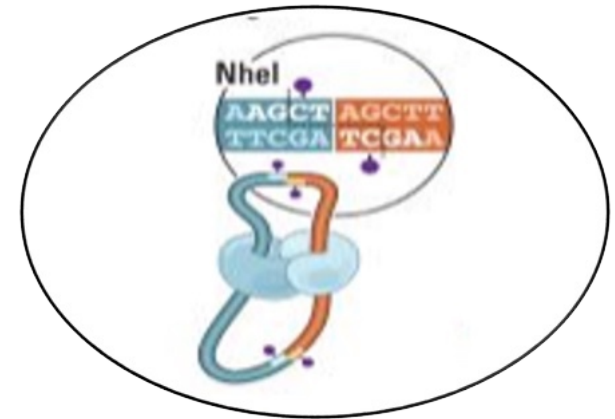
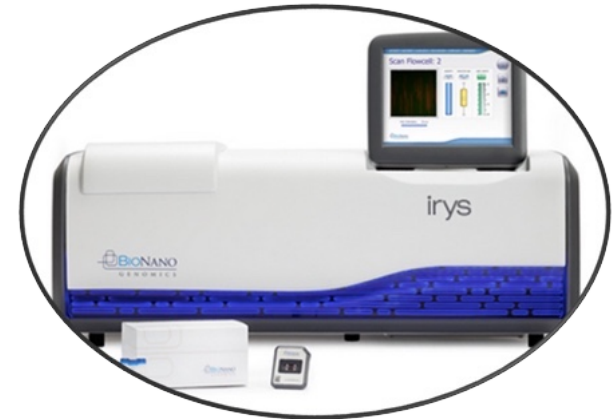
Sample Preparation

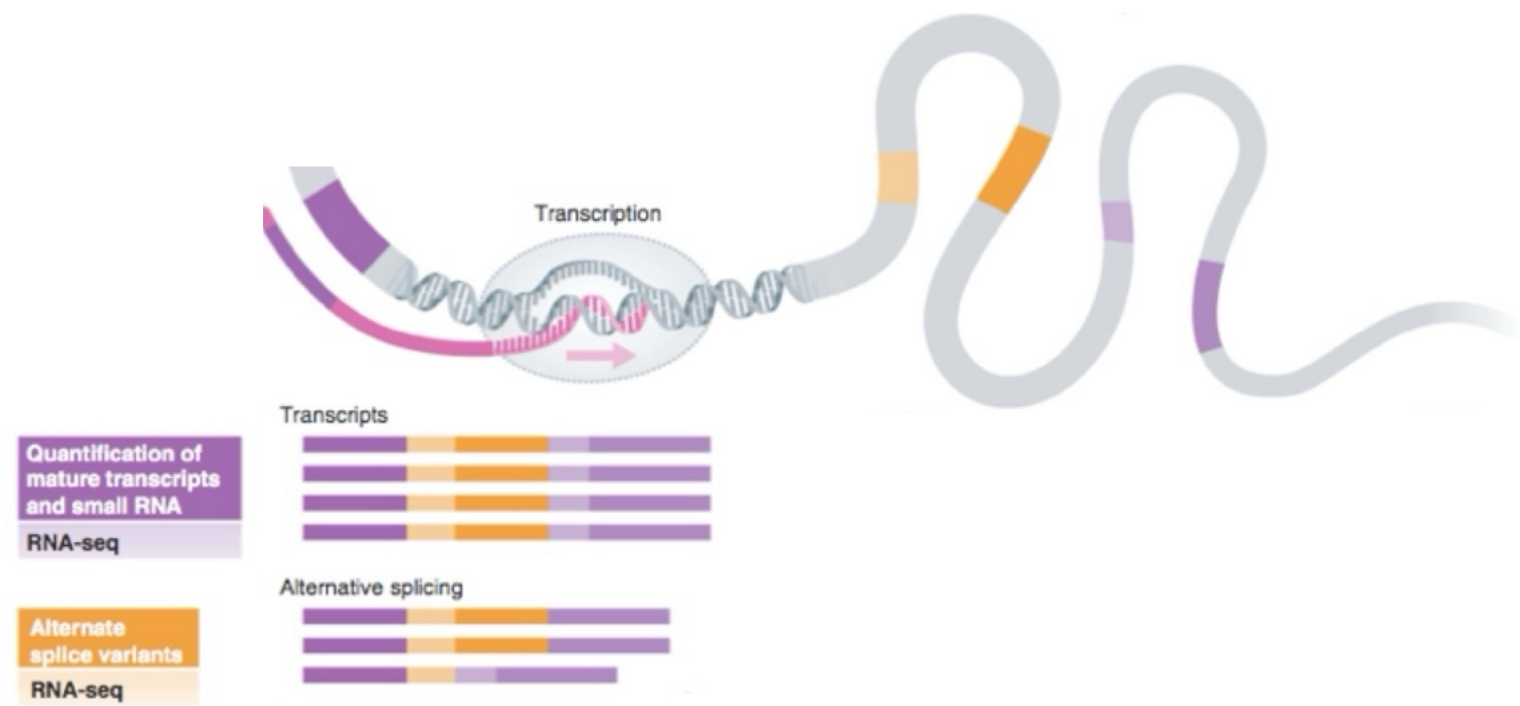


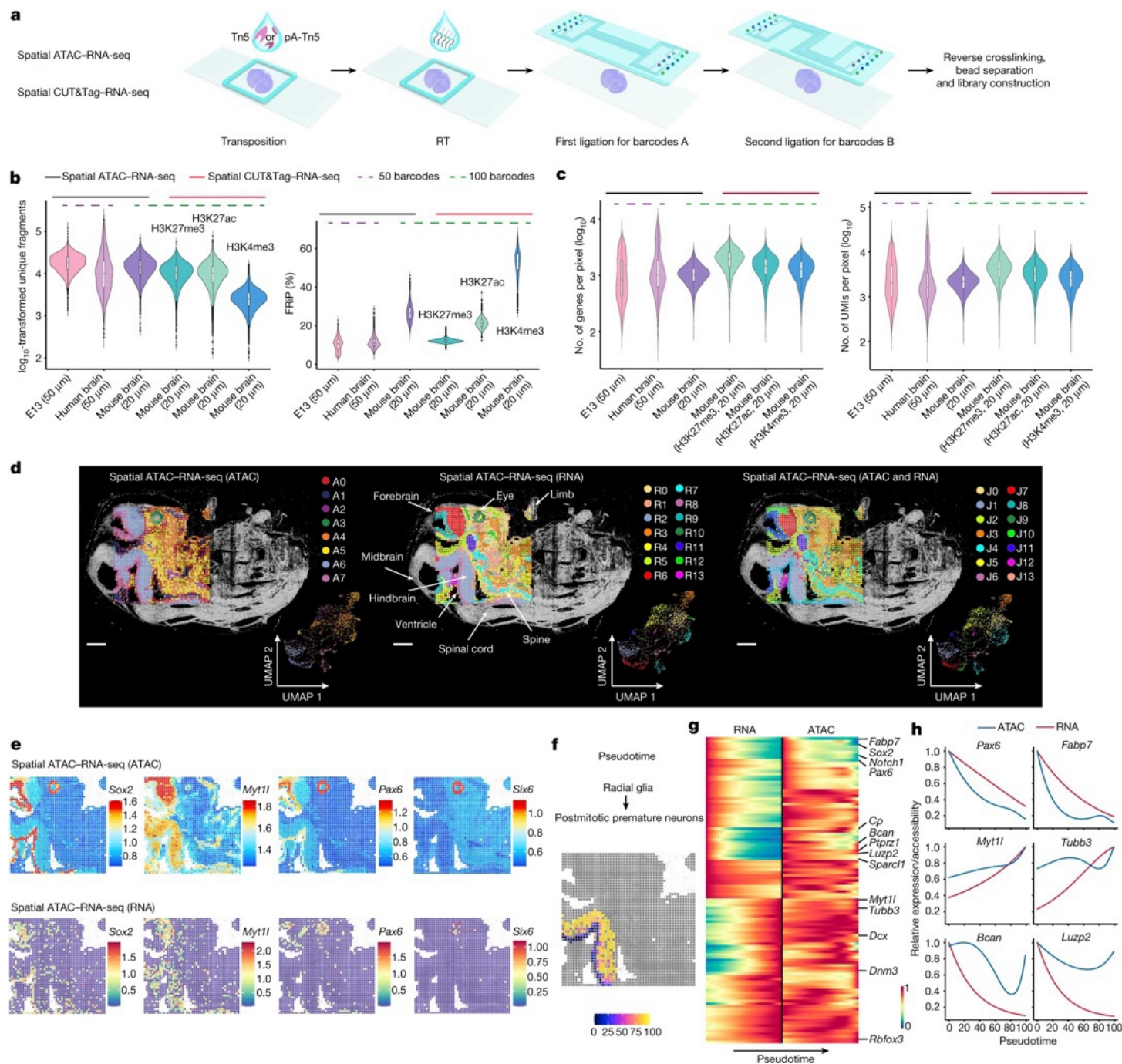
Sequencing



Chromosome Mapping



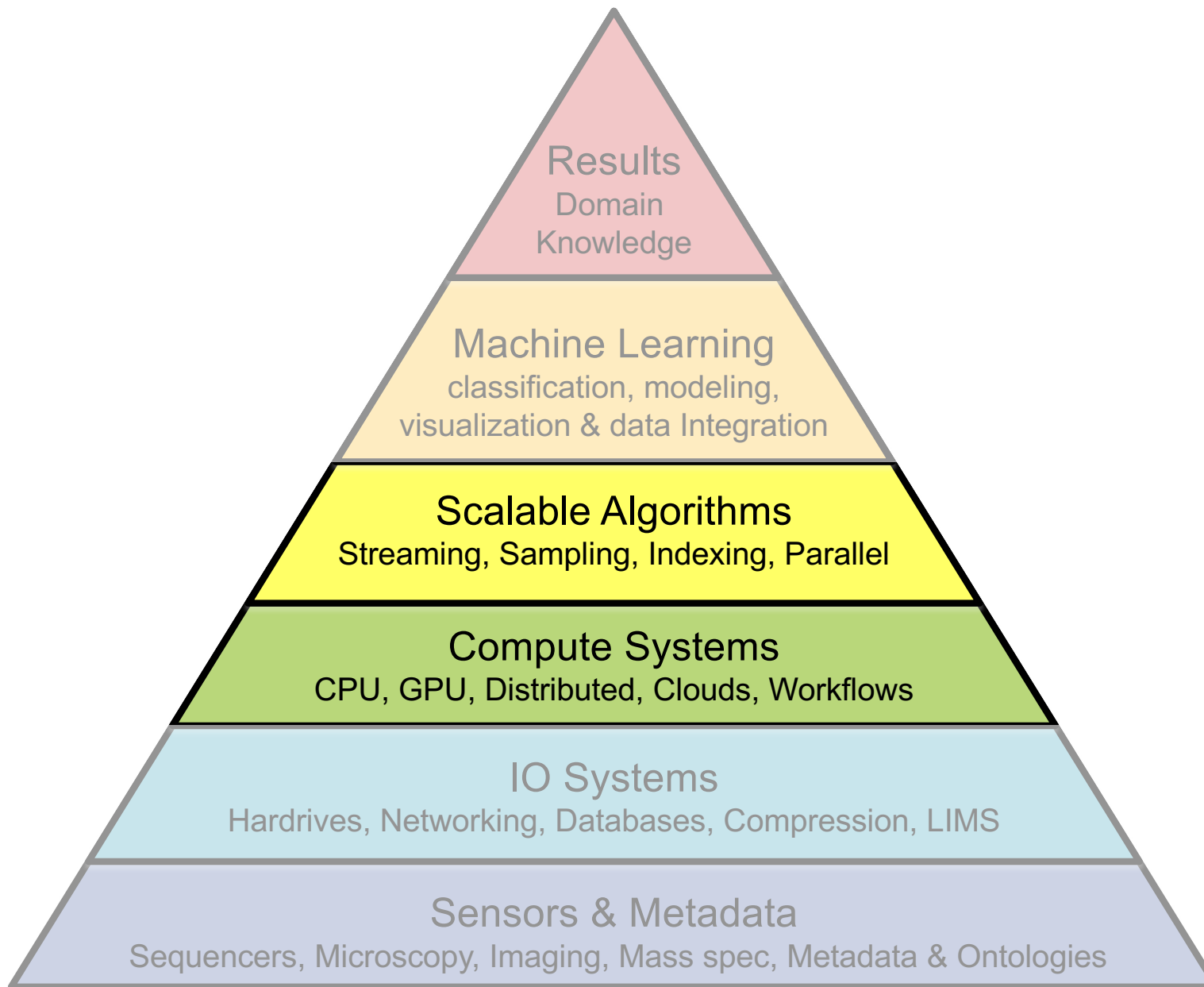




Spatial epigenome–transcriptome co-profiling of mammalian tissues

Zhang et al. (2023) *Nature*. <https://doi.org/10.1038/s41586-023-05795-1>

Applied Genomics

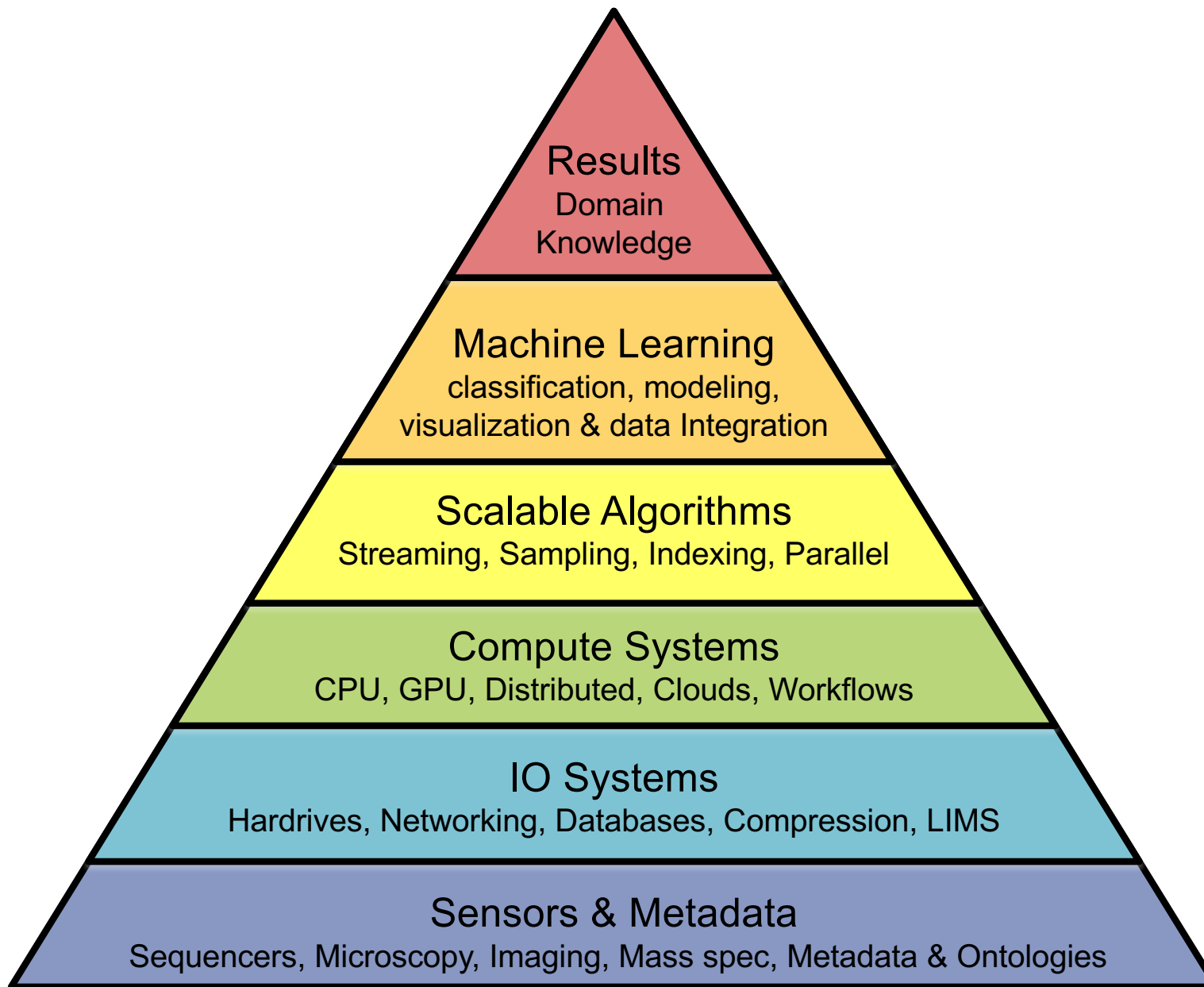


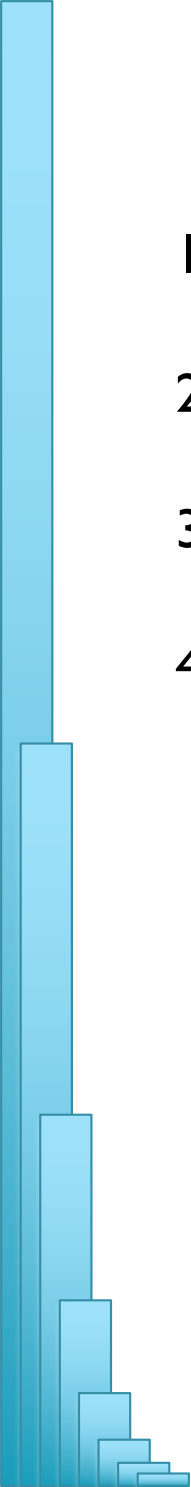
Potential Topics

- Genome assembly, whole genome alignment
- Full text indexing: Suffix Trees, Suffix Arrays, FM-index
- Dynamic Programming: Edit Distance, sequence similarity
- Read mapping & Variant identification
- Gene Finding: HMMs, Plane-sweep algorithms
- RNA-seq: mapping, assembly, quantification
- ChIP-seq: Peak finding, motif finding
- Methylation-seq: Mapping, CpG island detection
- HiC: Domain identification, scaffolding
- Chromatin state analysis: ChromHMM, Enformer
- Scalable genomics: Cloud computing, scalable data structures
- Population & single cell analysis: clustering, pseudotime
- Disease analysis, cancer genomics, Metagenomics
- Deep learning in genomics



Applied Genomics





Next Steps

1. Reflect on the magic and power of DNA 😊
2. Check out the course webpage
3. Register on Piazza
4. Get Ready for assignment I
 1. Set up conda
 2. Set up Dropbox for yourself!
 3. Get comfortable on the command line